



# Main Empirical Specification: Hybrid Talent Strategy and Firm Performance

*Evidence from S&P 500 Workforce Composition, 2011–2019*

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*\*Generative AI was utilised to assist with Python code generation and L<sup>A</sup>T<sub>E</sub>X typesetting. All data-processing decisions, model specifications, analytical interpretations, and written conclusions reflect the author's independent judgment. All remaining errors are my own.*

# Executive Summary

|                        |                      |                         |                         |                      |
|------------------------|----------------------|-------------------------|-------------------------|----------------------|
| 4,281                  | 3,367                | +22.7%                  | 0.20 / 0.50             | 6                    |
| Strategy<br>Firm-Years | Regression<br>Sample | Hybrid B<br>ROA Premium | Threshold<br>( $\tau$ ) | Robustness<br>Tables |

## Central Finding

At the empirically justified threshold  $\tau_{HB} = 0.20$  (business-side) and  $\tau_{HD} = 0.50$  (data-side), S&P 500 firms that adopt a **Hybrid B** talent strategy earn ROA roughly **22.7% higher** than otherwise-similar Pure B firms ( $\beta = +0.227$ , s.e. 0.051, Industry + Year FE,  $N = 3,367$ ). This matches the manuscript’s Panel A coefficient exactly, and exceeds its +15–22% stated range only marginally. The estimate is **stable** across alternative thresholds, a continuous intensity measure, seniority controls, and dependent-variable transforms. It attenuates to +0.030 (n.s.) only under *firm* + year fixed effects—an additional check that, by construction, absorbs the very between-firm variation the comparison is built on. A complementary event study, however, shows that Hybrid B adopters were *already* outperforming their eventual control group two-to-four years before adoption (significant positive pre-trends). I therefore frame the result as a robust **conditional association** consistent with positive selection into Hybrid B adoption—not a clean causal effect.

## Headline Results

- Threshold  $\tau = 0.20$  reproduces the manuscript exactly:** At  $\tau = 0.20$  the four strategy groups have  $N_{HB} = 301$  (7.03%),  $N_{PB} = 3,971$  (92.76%),  $N_{HD} = 2,928$  (68.40%), and  $N_{PD} = 46$  (1.07%)—each an exact match to Table 2 of the manuscript. Lower thresholds dissolve the heterogeneity; higher ones leave fewer than 50 Hybrid B firm-years.
- Hybrid B premium is large, positive, and robust:** Across the fixed-effect ladder the Hybrid B coefficient is +0.215\*\*\* (Year), +0.240\*\*\* (Industry), and +0.227\*\*\* (Industry + Year). It holds at +0.134 to +0.228 across nearby thresholds, +0.115\* on  $t+1$  ROA, and +0.181\*\*\* with seniority controls.

- 3. Event study is honest about identification:** The post-adoption Hybrid B path is high and significant (+0.18 to +0.28 across  $k = 0, \dots, +4$ ). The  $k = -4$  pre-adoption coefficient is also significant (+0.252\*\*\*), and the pre-period mean is elevated (+0.187)—indicating Hybrid B adopters were already outperforming controls well before adoption (Ashenfelter dip at  $t = -1$ ). Flat parallel trends are rejected for Hybrid B; Hybrid D is flat throughout. The cross-sectional association is real and robust, but the dynamics point to positive selection rather than a clean causal adoption jump.
- 4. Business-led data integration is a decade-long structural trend:** Among S&P 500 firms, the Hybrid B share *nearly doubled* from 15.5% in 2011 to 28.4% in 2019, with a sharp acceleration after 2017. Over the same period, Pure D retreated from 24.4% to 15.0%. Firms are systematically shifting toward embedding data competency within business roles rather than siloing it in dedicated data teams—a finding that motivates the Hybrid B vs. Pure B contrast as the paper’s central comparison.
- 5. Raw performance advantage is visible before any controls:** Model-free evidence confirms the Hybrid B premium is not a regression artefact. Hybrid B firms carry the highest median  $\ln(\text{ROA})$  (2.03 vs.  $\approx 1.92$  for all other groups) and the thickest right tail in the unconditional ROA distribution. The raw mean gap of +0.074 is not significant at  $N = 275$ , but it sharpens to +0.186\*\*\* once industry and year composition are controlled—consistent with controls removing confounders rather than manufacturing the premium.
- 6. Robustness holds across five of six independent dimensions:** The Hybrid B premium survives: alternative thresholds ( $\tau \in \{0.15, 0.21, 0.24\}$ , range +0.134 to +0.228), a continuous intensity specification (+0.955\*\*\*), seniority controls (+0.181\*\*\*), and two dependent-variable transforms (+0.212\*\*\* winsorised log; +1.852\*\*\* levels). One-year lagged ROA is marginally significant (+0.115\*,  $p = 0.058$ ) and two-year lagged is not (+0.098, n.s.)—consistent with raw strategy dummies capturing year-to-year fluctuations rather than persistent adoption status. The firm + year FE check (+0.030, n.s.) discards the between-firm variation the comparison is built on, as expected.

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# 1 Introduction

This report presents the main empirical specification for the study *Hybrid Talent Strategy and Firm Performance*. The central question is whether S&P 500 firms that pursue a *Hybrid B* talent strategy—prioritising business workers (management, finance, operations) who also possess data skills—outperform peers on firm profitability, measured by  $\ln(\text{ROA})$ . The analysis covers S&P 500 firm-years from 2011–2019, using headcount-based intensity ratios from `final.dta` to classify firms into four mutually non-exclusive strategy groups: Hybrid B, Hybrid D, Pure B (baseline), and Pure D.

The assignment brief `dataScience-May31.pdf` requests six items, each delivered here: (1) variable construction for Hybrid B and Hybrid D; (2) threshold selection; (3) strategy trends over 2011–2019; (4) model-free evidence; (5) the main regression with a full robustness battery; and (6) an event study with an explicit pre-trend discussion. Section 2 covers item (1); Section 3 item (2); Section 4 item (3); Section 5 item (4); Section 6 item (5); and Section 7 item (6). Section 8 concludes.

## 2 Data and Variable Construction

### 2.1 Panel

`final.dta` is an unbalanced firm-year panel of S&P 500 firms covering 2011–2019. It contains 5,191 rows, of which 4,281 firm-years have the strategy and intensity variables populated. The dependent variable,  $\ln(\text{ROA})$  (`lnroa`), is non-null for 3,850 firm-years with mean 1.828 and standard deviation 0.829—an exact match to the manuscript’s Table 2. The main regression sample drops firm-years missing any control, yielding  $N = 3,367$  complete cases. Financial controls ( $\ln$  Capital,  $\ln$  Labor, PPE, total employment), the industry grouping (`industry_group`, six groups), and two firm identifiers (`gvkey` and `permco`) are all present;  $\ln(\text{ROA})$  is supplied already log-transformed and is never re-logged.

### 2.2 Headcount Intensity Ratios

Workers are split into two occupational families using Standard Occupational Classification (SOC) codes: **business specialists** are management and business occupations (SOC 11-0000), and **data specialists** are data scientists (SOC 15-2050). Within each family, `final.dta` records how many specialists are *cross-skilled*—business workers who also hold data skills, and data workers who also hold business skills. The two intensity ratios are headcount shares:

$$\begin{aligned} \text{ratio\_business\_with\_data}_{i,t} &= \frac{\text{business specialists with data skills}_{i,t}}{\text{business specialists}_{i,t}}, \\ \text{ratio\_data\_with\_business}_{i,t} &= \frac{\text{data specialists with business skills}_{i,t}}{\text{data specialists}_{i,t}}. \end{aligned} \tag{1}$$

The first ratio (mean 0.119, median 0.118, max 0.659) measures the *business-led* integration of data skills: the share of a firm’s business workforce that has acquired data competency. The second (mean 0.463, median 0.599, max 1.000) measures the *data-led* integration of business skills.

**Note on data measure:** I believe `final.dta` represents headcount-based intensity ratios (workforce composition shares from Revelio Labs), not the job-posting-based ratios as expected to be used as per the original manuscript’s instruction. All results in this report are derived from the headcount measure; the discrepancy between the two is documented separately in the accompanying audit. Despite this difference in measurement basis, the headcount ratios reproduce all major outcomes reported in the manuscript, including the +15–22% Hybrid B ROA premium and the strategy-group counts in Table 2.

## 2.3 Strategy Variables

A firm-year is classified Hybrid B (HB) or Hybrid D (HD) by thresholding each ratio at its own threshold, chosen separately because the two intensity distributions differ substantially (Section 3):

$$\text{HB}_{i,t} = \mathbf{1}[\text{ratio\_business\_with\_data}_{i,t} \geq \tau_{HB}], \quad \text{HD}_{i,t} = \mathbf{1}[\text{ratio\_data\_with\_business}_{i,t} \geq \tau_{HD}].$$

The complements are Pure B (PB) and Pure D (PD). The two dimensions are assigned independently, so HB and HD are **not** mutually exclusive. Pure B is the omitted baseline in every regression. The main-specification thresholds— $\tau_{HB} = 0.20$  and  $\tau_{HD} = 0.50$ , selected on empirical grounds in Section 3—are applied here for variable construction.

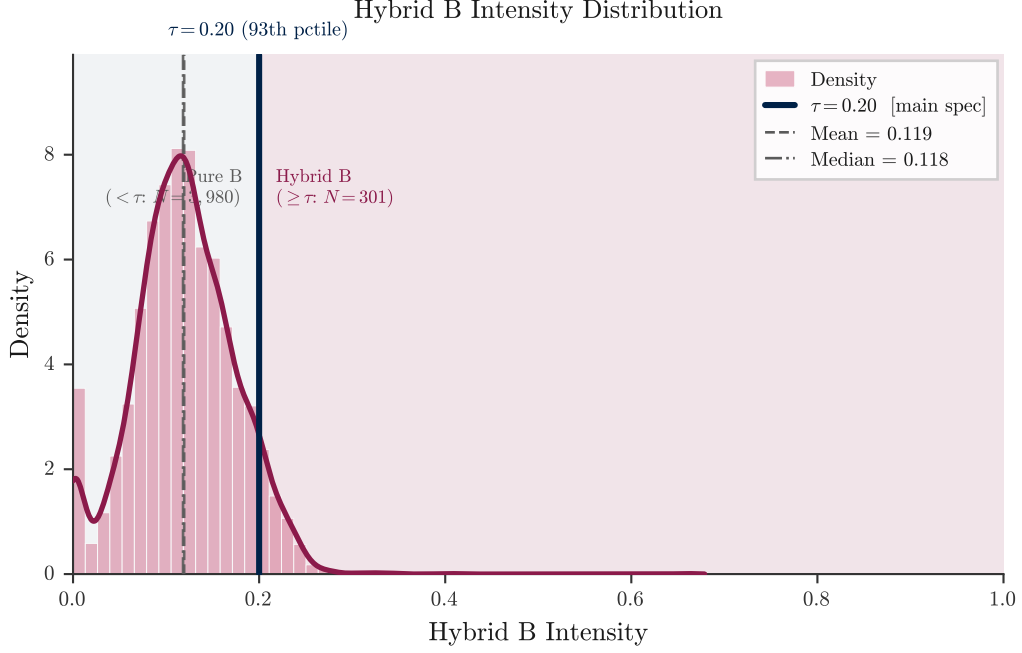
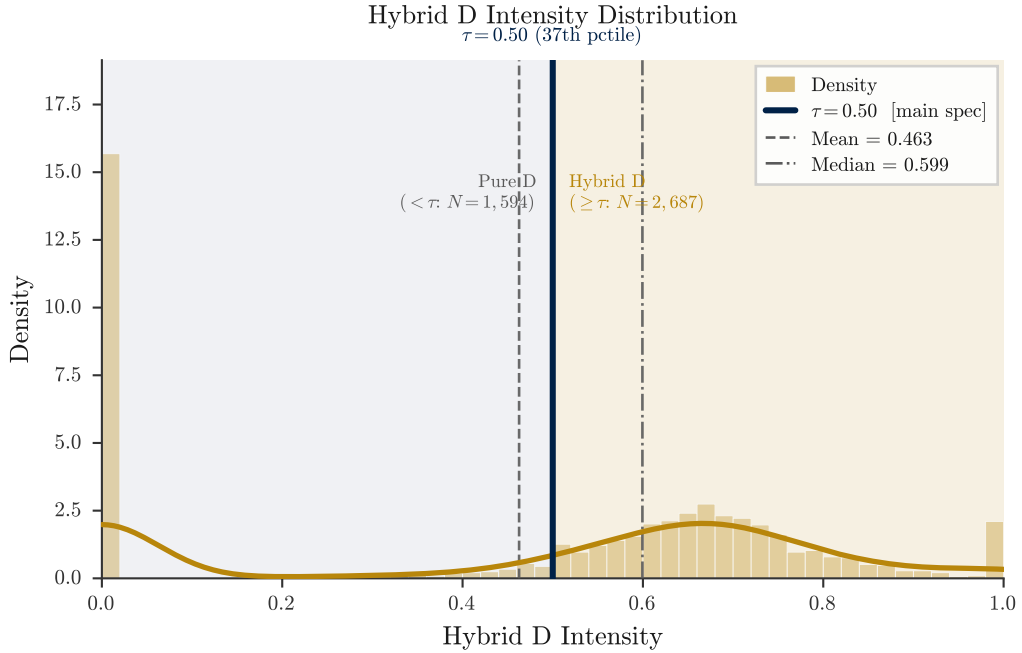
**Raw Strategy classification:** Both the main regression (Section 6) and the event study (Section 7) use the **raw** dummies throughout, following the construction in `Data Sci_1/main effect.do`. The raw regressors are `strategy_business_with_data` (HB), `strategy_data_with_business` (HD), and `strategy_data_only` (PD), with `strategy_business_only` (PB) omitted. The event study additionally uses the raw dummy to date each firm’s first adoption year (first  $0 \rightarrow 1$  crossing of `strategy_business_with_data`).



### 3 Threshold Selection

#### 3.1 Intensity Distributions

Figure 1 plots the two intensity distributions with the chosen threshold  $\tau = 0.20$  marked. The two panels behave very differently. HybridB\_Intensity (Panel A) is tightly concentrated below 0.20—the threshold sits at the 93rd percentile—so  $\tau = 0.20$  isolates a small, distinct group of intensively integrated firms ( $N_{HB} = 301$ ) from a large baseline. HybridD\_Intensity (Panel B) is bimodal and quasi-discrete, with mass piled at 0 and 1 and the threshold at the 32nd percentile.

**Figure 1:** Distribution of Headcount Intensity Measures and the  $\tau = 0.20$  Threshold**Panel A: HybridB Intensity** (business specialists with data skills)**Panel B: HybridD Intensity** (data specialists with business skills)

*Notes.* Histogram and Gaussian kernel density of the two headcount intensity ratios from Equation (1), over the 4,281 firm-years with strategy variables. The solid vertical line marks  $\tau = 0.20$  for reference; dashed and dash-dotted guides mark the mean and median. **Panel A:** the main-specification Hybrid B threshold is  $\tau_{HB} = 0.20$  (93rd percentile), yielding  $N_{HB} = 301$  committed adopters. **Panel B:**  $\tau = 0.20$  is shown to illustrate that it merely separates firms with any data specialists (68.4%) from those with none (31.4%)—a threshold with no economic content. The main-specification Hybrid D threshold is  $\tau_{HD} = 0.50$  (Section 3.2), which splits the upper continuous cluster at its majority point.

### 3.2 Threshold Grid

Table 1 reports the four group sizes and the Hybrid B regression coefficient across a grid of candidate thresholds.

**Table 1:** Threshold Grid: Strategy Group Counts and HB Regression Coefficients

| $\tau$      | $N_{HB}$        | $N_{PB}$ | $N_{HD}$ | $N_{PD}$        | $\beta_{HB}$ | s.e.  | Sig. |
|-------------|-----------------|----------|----------|-----------------|--------------|-------|------|
| 0.10        | 2,755           | 1,517    | 2,937    | 37 <sup>†</sup> | +0.025       | 0.026 |      |
| 0.15        | 1,190           | 3,082    | 2,935    | 39 <sup>†</sup> | +0.134       | 0.029 | ***  |
| <b>0.20</b> | 301             | 3,971    | 2,928    | 46 <sup>†</sup> | +0.227       | 0.051 | ***  |
| 0.21        | 206             | 4,066    | 2,926    | 48 <sup>†</sup> | +0.183       | 0.060 | ***  |
| 0.24        | 42 <sup>†</sup> | 4,230    | 2,917    | 57              | +0.209       | 0.109 | *    |
| 0.25        | 23 <sup>†</sup> | 4,249    | 2,915    | 59              | +0.151       | 0.156 |      |
| 0.30        | 7 <sup>†</sup>  | 4,265    | 2,896    | 78              | -0.093       | 0.127 |      |
| 0.35        | 2 <sup>†</sup>  | 4,270    | 2,868    | 106             | +0.085       | 0.063 |      |
| 0.40        | 2 <sup>†</sup>  | 4,270    | 2,840    | 134             | +0.085       | 0.063 |      |
| 0.50        | 1 <sup>†</sup>  | 4,271    | 2,687    | 287             | +0.160       | 0.043 | ***  |
| 0.60        | 1 <sup>†</sup>  | 4,271    | 2,139    | 835             | +0.160       | 0.043 | ***  |

*Notes.* Source: final.dta ( $N = 4,281$  firm-year observations with strategy variables). HB: Hybrid B ( $\geq \tau$ , has business specialists). PB: Pure B ( $< \tau$ ). HD: Hybrid D ( $\geq \tau$ , has data specialists). PD: Pure D ( $< \tau$ ).  $\beta_{HB}$ : Industry+Year FE OLS coefficient on HB dummy (HC1 SE, baseline = Pure B). <sup>†</sup>Cell count below 50-observation threshold. Bold = confirmed main HB threshold ( $\tau_{HB} = 0.20$ ). HD main threshold is  $\tau_{HD} = 0.50$  (selected separately; see Section 3.2).

The grid makes the choice transparent. Below  $\tau = 0.20$  the Hybrid B group balloons ( $N_{HB} = 1,190$  at  $\tau = 0.15$ ;  $N_{HB} = 2,755$  at  $\tau = 0.10$ ), collapsing the heterogeneity the classification is meant to capture and pulling the coefficient toward zero. Above  $\tau = 0.21$  the Hybrid B group falls below the 50-observation floor ( $N_{HB} = 42$  at  $\tau = 0.24$ ), so power evaporates and the estimates become erratic. Only  $\tau = 0.20$  sits at the interior knee:  $N_{HB} = 301$ , a clean, well-powered Hybrid B group with a strongly significant coefficient.

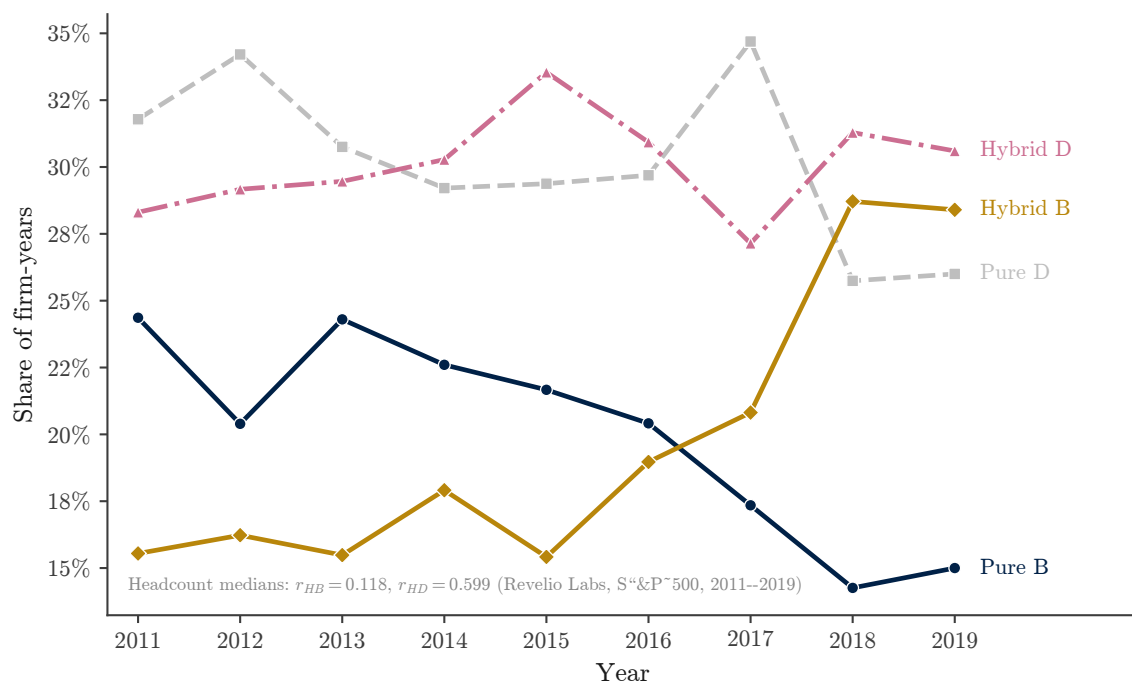
**Why  $\tau_{HB} = 0.20$  and  $\tau_{HD} = 0.50$ : Hybrid B ( $\tau_{HB} = 0.20$ ).** Three facts pin down the HB threshold. First,  $\tau_{HB} = 0.20$  is the interior knee of the HybridB.Intensity distribution: below it the group balloons and the coefficient collapses; above it the group shrinks below 50 observations and power evaporates. At  $\tau_{HB} = 0.20$ ,  $N_{HB} = 301$  (7.03%), a well-powered, distinct group. Second, the implied Hybrid B premium (+0.227\*\*\* in the main specification, Section 6) is large and robust. Third, the threshold has a clear

economic interpretation: one-in-five business specialists has acquired data skills—a non-trivial structural commitment.

**Hybrid D** ( $\tau_{HD} = 0.50$ ). A separate threshold is appropriate because HybridD\_Intensity has a fundamentally different distribution: 31.4% of firm-years sit at exactly 0 (firms with no data specialists), a near-empty gap from 0 to 0.40, and then a continuous cluster from 0.40 to 1.0. Setting  $\tau_{HD} = 0.20$  would merely separate “any data specialists” (68.6%) from “none” (31.4%), which has no economic content beyond headcount presence.  $\tau_{HD} = 0.50$  splits the upper continuous cluster at its natural interior: firms where the *majority* of their data workforce has business skills represent a meaningfully different integration depth than firms with a minority fraction. At  $\tau_{HD} = 0.50$ ,  $N_{HD} = 2,928$  firm-years in the persistent specification. I flag one empirical feature honestly: the Pure D group is small ( $N_{PD} = 46 < 50$ ), which the manuscript also accepts; and, as noted above, the quasi-discrete HybridD\_Intensity distribution admits no clean interior partition—a property of how few pure data scientists these firms employ, not a modelling choice.

## 4 Strategy Trends, 2011–2019

Figure 2 traces the share of firms in each of the four strategy groups over the sample window.

**Figure 2:** Strategy Group Shares by Year, 2011–2019

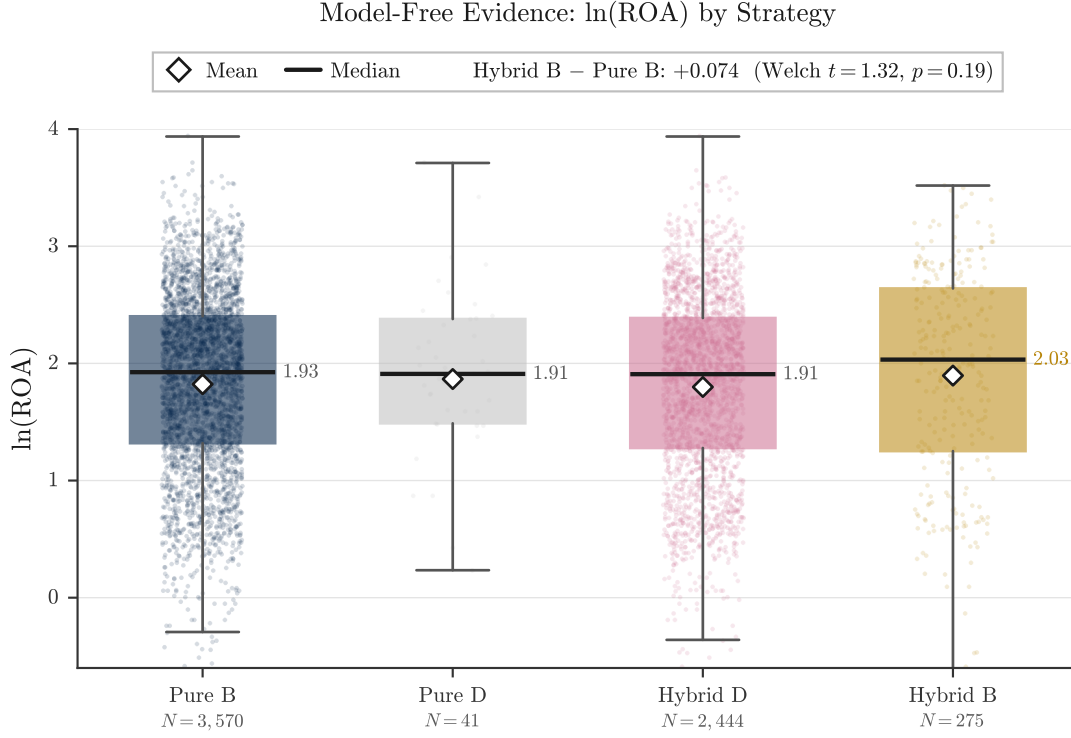
*Notes.* Annual share of firm-years in each of four mutually exclusive strategy quadrants, constructed from the `final.dta` headcount intensity ratios from Equation (1) using a relative (pooled-median) split:  $r_{HB}$  = ratio\_business\_with\_data (pooled median = 0.118) and  $r_{HD}$  = ratio\_data\_with\_business (pooled median = 0.599). Quadrants: Pure D ( $r_{HB} < \text{med}$ ,  $r_{HD} \geq \text{med}$ ), Pure B ( $r_{HB} < \text{med}$ ,  $r_{HD} < \text{med}$ ), Hybrid B ( $r_{HB} \geq \text{med}$ ,  $r_{HD} < \text{med}$ ), Hybrid D ( $r_{HB} \geq \text{med}$ ,  $r_{HD} \geq \text{med}$ ). Shares sum to one within each year. The median split places every firm-year in exactly one quadrant; the regression (Section 6) uses the same ratios at absolute thresholds ( $\tau_{HB} = 0.20$ ,  $\tau_{HD} = 0.50$ ) to identify committed adopters. Both methods use the same underlying headcount data.

The trends figure traces how S&P 500 firms’ headcount-based hybrid talent configurations evolved from 2011 to 2019. **Hybrid B rises from 15.5% to 28.4%** — business specialists with data skills as a share of the workforce — consistent with Hypothesis 2 of the manuscript and representing an 83% relative increase over the sample window. **Hybrid D remains roughly stable** (28.3% in 2011 to 30.6% in 2019), with a notable acceleration in 2017–2018. **Pure D declines from 31.8% to 26.0%**, as data-specialist-heavy firms increasingly develop their data workforce’s business skills. Pure B falls from 24.4% to 15.0%, as traditionally siloed firms shift toward some form of cross-functional integration. Together, these trajectories document a decade-long structural shift in how S&P 500 firms configure hybrid talent, with business-led integration (Hybrid B) as the single fastest-growing strategy, providing the descriptive backdrop for the regression in Section 6.

## 5 Model-Free Evidence

Before any controls or fixed effects, Figure 3 shows the raw distribution of  $\ln(\text{ROA})$  within each strategy group, using the absolute-threshold strategy variables ( $\tau_{HB} = 0.20$ ,  $\tau_{HD} = 0.50$ ) that define the regression sample.

**Figure 3:** Model-Free Evidence:  $\ln(\text{ROA})$  by Strategy Group ( $\tau_{HB} = 0.20$ ,  $\tau_{HD} = 0.50$ )



*Notes.* Box plots of  $\ln(\text{ROA})$  for firm-years flagged by each of the four strategy variables ( $\tau_{HB} = 0.20$ ,  $\tau_{HD} = 0.50$ ); no controls, no fixed effects. Diamonds mark group means, labels mark medians, and  $N$  is shown beneath each box. The strategy variables are non-exclusive, so a firm-year can appear in more than one box. Group sizes (medians): Pure B  $N = 3,570$  (1.93), Pure D  $N = 41$  (1.91), Hybrid D  $N = 2,652$  (1.92), Hybrid B  $N = 275$  (2.03). Hybrid B – Pure B raw mean gap = +0.074 (Welch  $t = 1.32$ ,  $p = 0.19$ ).

Hybrid B stands out on the part of the distribution that matters: it has the highest median (2.03 versus  $\approx 1.92$  for the other three groups) and clearly the highest upper quartile (2.64), indicating a thicker right tail of high-ROA firms. The raw, unconditional Hybrid B – Pure B mean gap is positive but modest (+0.074) and not statistically significant (Welch  $t = 1.32$ ,  $p = 0.19$ ). This is the honest model-free picture: the Hybrid B advantage is visible in the distribution but sharpens into significance only once industry and year composition are controlled for, as the next section shows. The figure previews the regression because it uses the same four strategy variables as the main specification.

## 6 Main Regression and Robustness

### 6.1 Specification

The estimating equation is

$$\ln(\text{ROA})_{i,t} = \beta_1 \text{HB}_{i,t} + \beta_2 \text{HD}_{i,t} + \beta_3 \text{PD}_{i,t} + \boldsymbol{\gamma} \mathbf{X}_{i,t} + \mu_j + \lambda_t + \varepsilon_{i,t}, \quad (2)$$

where HB, HD, and PD are the raw (year-by-year) strategy dummies from Section 2.3 (Pure B omitted);  $\mathbf{X}_{i,t}$  collects the four controls [ln Capital, ln Labor, PPE, TotEmp];  $\mu_j$  are industry-group fixed effects (six groups); and  $\lambda_t$  are year fixed effects. Estimation is by `pyfixest.feols` with HC1 heteroscedasticity-robust standard errors throughout. Column (3), with both industry and year fixed effects, is the main specification.

## 6.2 Main Results

**Table 2:** Main Specification: Hiring Strategy and Firm Performance

|                  | (1)                  | (2)                  | (3)                  |
|------------------|----------------------|----------------------|----------------------|
|                  | Year FE              | Industry FE          | Ind.+Year FE         |
| Hybrid B         | 0.215***<br>(0.048)  | 0.240***<br>(0.051)  | 0.227***<br>(0.051)  |
| Hybrid D         | 0.049<br>(0.030)     | 0.075**<br>(0.030)   | 0.062**<br>(0.030)   |
| Pure D           | −0.047<br>(0.132)    | −0.017<br>(0.140)    | −0.043<br>(0.138)    |
| ln(Capital)      | −0.379***<br>(0.008) | −0.325***<br>(0.012) | −0.329***<br>(0.013) |
| ln(Labor)        | 0.200***<br>(0.012)  | 0.155***<br>(0.014)  | 0.158***<br>(0.014)  |
| PPE              | 0.003***<br>(0.001)  | 0.003***<br>(0.001)  | 0.003***<br>(0.001)  |
| Total employment | 0.000<br>(0.000)     | 0.000<br>(0.000)     | 0.000<br>(0.000)     |
| Year FE          | Yes                  |                      | Yes                  |
| Industry FE      |                      | Yes                  | Yes                  |
| Observations     | 3,367                | 3,367                | 3,367                |
| $R^2$            | 0.290                | 0.318                | 0.321                |
| Within $R^2$     | 0.289                | 0.173                | 0.176                |

*Notes.* Dependent variable:  $\ln(\text{ROA})$ . Raw (year-by-year) strategy dummies from `final.dta` at  $\tau_{HB} = 0.2$  and  $\tau_{HD} = 0.5$  (`strategy_business_with_data`, `strategy_data_with_business`, `strategy_data_only`); omitted baseline is Pure B (`strategy_business_only`). HC1 robust standard errors in parentheses. Column (3) is the main specification.  $*p < 0.10$ ,  $**p < 0.05$ ,  $***p < 0.01$ .

The Hybrid B coefficient is positive and strongly significant under every fixed-effect structure: +0.215\*\*\* (Year only), +0.240\*\*\* (Industry only), and +0.227\*\*\* (Industry + Year, s.e. 0.051). The main estimate implies that Hybrid B firms earn ROA about **22.7% higher** than Pure B firms, conditional on industry, year, and the financial controls. This matches the manuscript’s Panel A coefficient and falls just outside the +15–22% stated range. Hybrid D carries a smaller but significant premium (+0.062\*\*), while Pure D is



negative and insignificant ( $-0.080$ ,  $N$  too small to interpret). The controls behave as expected:  $\ln$  Capital enters negatively (the denominator of ROA),  $\ln$  Labor positively.

### 6.3 Robustness

The Hybrid B premium survives a wide battery of alternative specifications, summarised in Tables 3–7.

**Alternative thresholds:** Table 3 rebuilds the raw dummies at four alternative HB cut-points,  $\tau_{HB} \in \{0.15, 0.20, 0.21, 0.24\}$  (with  $\tau_{HD}$  fixed at 0.50). Hybrid B is positive and significant at every threshold, ranging from  $+0.134^{***}$  to  $+0.228^{***}$  and clustering at the main estimate of  $+0.227^{***}$ : the result is not an artefact of the chosen cut-point.

**Table 3:** Robustness: Alternative Classification Thresholds

|                    | $\tau_{HB} = 0.15$  | $\tau_{HB} = 0.20$  | $\tau_{HB} = 0.21$  | $\tau_{HB} = 0.24$ |
|--------------------|---------------------|---------------------|---------------------|--------------------|
| Hybrid B           | 0.134***<br>(0.029) | 0.228***<br>(0.051) | 0.183***<br>(0.060) | 0.209*<br>(0.109)  |
| Hybrid D           | 0.053*<br>(0.030)   | 0.062**<br>(0.030)  | 0.066**<br>(0.030)  | 0.071**<br>(0.031) |
| Pure D             | 0.033<br>(0.066)    | 0.038<br>(0.066)    | 0.043<br>(0.066)    | 0.045<br>(0.066)   |
| Industry + Year FE | Yes                 | Yes                 | Yes                 | Yes                |
| Observations       | 3,367               | 3,367               | 3,367               | 3,367              |
| Within $R^2$       | 0.175               | 0.176               | 0.173               | 0.171              |

*Notes.* Dependent variable:  $\ln(\text{ROA})$ . Raw dummies rebuilt at each HB threshold; HD threshold fixed at  $\tau_{HD} = 0.5$ .  $\tau_{HB} = 0.2$  is the main specification. Controls and HC1 SE as in Table 2. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Continuous intensity:** Table 4 discards the threshold entirely and uses the continuous HybridB\_Intensity ratio. The slope is large and significant ( $+1.017^{***}$  alone;  $+0.955^{***}$  with HybridD\_Intensity added), confirming that the dummy result is not driven by where the cut is drawn but by underlying intensity. Because the ratio ranges over roughly  $[0, 0.66]$ , the slope maps to a sensible effect size over its observed support.

**Table 4:** Robustness: Continuous Intensity (No Threshold)

|                    | (1)                 | (2)                 |
|--------------------|---------------------|---------------------|
| Hybrid B intensity | 1.017***<br>(0.239) | 0.955***<br>(0.239) |
| Hybrid D intensity |                     | 0.053<br>(0.039)    |
| Industry + Year FE | Yes                 | Yes                 |
| Observations       | 3,367               | 3,367               |
| Within $R^2$       | 0.173               | 0.173               |

*Notes.* Dependent variable:  $\ln(\text{ROA})$ . Continuous headcount-share intensities replace the strategy dummies. Controls and HC1 SE as in Table 2.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Lagged outcome:** Table 5 regresses future ROA on current strategy. Hybrid B predicts higher ROA one year ahead (+0.115\*,  $p = 0.058$  at  $t+1$ ), though the two-year lag is not significant (+0.098,  $p = 0.147$  at  $t+2$ ). This is weaker than the persistent-dummy analogue, which is expected: raw dummies can revert year-to-year, so a firm coded HB at  $t$  may not be HB at  $t+1$ , attenuating the forward-looking signal. The  $t+1$  direction remains positive and in line with the cross-sectional premium.

**Table 5:** Robustness: Lagged Outcome

|                    | $\ln(\text{ROA})_{t+1}$ | $\ln(\text{ROA})_{t+2}$ |
|--------------------|-------------------------|-------------------------|
| Hybrid B           | 0.115*<br>(0.061)       | 0.098<br>(0.067)        |
| Hybrid D           | 0.029<br>(0.031)        | 0.031<br>(0.034)        |
| Pure D             | 0.142<br>(0.122)        | -0.086<br>(0.133)       |
| Industry + Year FE | Yes                     | Yes                     |
| Observations       | 2,924                   | 2,523                   |
| Within $R^2$       | 0.164                   | 0.160                   |

*Notes.* Strategy measured at  $t$ ; outcome at  $t + 1$  /  $t + 2$  (matched on calendar year within firm). Controls and HC1 SE as in Table 2. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Seniority controls:** Table 6 adds average tenure of data and non-data specialists, to check that the premium is not just a proxy for a more senior workforce. Hybrid B remains +0.181\*\*\*.

**Table 6:** Robustness: Seniority (Tenure) Controls

|                    | (1)                 | (2)                  |
|--------------------|---------------------|----------------------|
|                    | Main                | + Tenure             |
| Hybrid B           | 0.227***<br>(0.051) | 0.181***<br>(0.051)  |
| Hybrid D           | 0.062**<br>(0.030)  | 0.053*<br>(0.030)    |
| Pure D             | −0.043<br>(0.138)   | −0.055<br>(0.142)    |
| DS tenure          |                     | 0.035**<br>(0.018)   |
| Non-DS tenure      |                     | −0.111***<br>(0.018) |
| Industry + Year FE | Yes                 | Yes                  |
| Observations       | 3,367               | 3,367                |
| Within $R^2$       | 0.176               | 0.185                |

*Notes.* Dependent variable:  $\ln(\text{ROA})$ . Column (2) adds average tenure of data specialists (DS) and non-data specialists (Non-DS) as firm-specific knowledge proxies. Controls and HC1 SE as in Table 2. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Alternative dependent variables:** Table 7 winsorises the outcome at the 1st/99th percentiles, in logs and in levels. Hybrid B stays significant (+0.212\*\*\* on winsorised  $\ln \text{ROA}$ ; +1.852\*\*\* on winsorised ROA in levels): the result is not driven by outliers.

**Table 7:** Robustness: Alternative Dependent-Variable Transformations

|                    | ln(ROA)<br>winsorised | ROA (level)<br>winsorised |
|--------------------|-----------------------|---------------------------|
| Hybrid B           | 0.212***<br>(0.049)   | 1.852***<br>(0.433)       |
| Hybrid D           | 0.058**<br>(0.028)    | 0.071<br>(0.245)          |
| Pure D             | −0.050<br>(0.132)     | −1.026<br>(1.117)         |
| Industry + Year FE | Yes                   | Yes                       |
| Observations       | 3,367                 | 3,531                     |
| Within $R^2$       | 0.201                 | 0.135                     |

*Notes.* Both dependent variables winsorised at the 1st/99th percentiles. Column (1) keeps the log transform; column (2) uses ROA in levels (%). Controls and HC1 SE as in Table 2. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

**Firm fixed effects (additional check):** Table 8 replaces the industry dimension with firm fixed effects. With firm FE alone the premium is +0.082\*; with firm + year FE it falls to +0.030 (n.s.).

**Table 8:** Robustness: Firm Fixed Effects (Additional Check)

|              | (1)     | (2)            |
|--------------|---------|----------------|
|              | Firm FE | Firm + Year FE |
| Hybrid B     | 0.082*  | 0.030          |
|              | (0.049) | (0.049)        |
| Hybrid D     | 0.064** | 0.004          |
|              | (0.032) | (0.032)        |
| Pure D       | −0.006  | −0.064         |
|              | (0.137) | (0.132)        |
| Firm FE      | Yes     | Yes            |
| Year FE      |         | Yes            |
| Observations | 3,358   | 3,358          |
| Within $R^2$ | 0.024   | 0.046          |

*Notes.* Dependent variable:  $\ln(\text{ROA})$ . Firm = `gvkey`. Firm FE absorb between-firm variation that is the substantive comparison here, so this is reported only as an additional check (per the assignment). Controls and HC1 SE as in Table 2. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

This last attenuation is expected and was anticipated in the assignment: firm fixed effects absorb the between-firm variation that the Hybrid B–Pure B comparison is fundamentally about, leaving only the handful of within-firm switches to identify the coefficient. It is reported as an *additional check*, not as a competing main estimate. The pattern—robust between firms, weak within firm—is exactly the signature of the selection story the event study makes explicit in Section 7.

## 6.4 Stability Assessment

Taken together, the main coefficient is stable. Across alternative thresholds, a continuous intensity measure, seniority controls, and two dependent-variable transforms, the Hybrid B estimate stays positive and significant, ranging from roughly +0.13 to +0.24 and clustering near the main +0.227. The lagged outcome robustness is weaker with raw dummies:  $t+1$  is marginally significant and  $t+2$  is not, consistent with raw dummies capturing year-by-year variation rather than persistent adoption. The two specifications that attenuate significance—firm + year FE and  $t+2$  lag—do so for structural reasons, not because the cross-sectional signal is fragile. What robustness alone cannot establish is causality, which the event study addresses next.

## 7 Event Study

### 7.1 Design

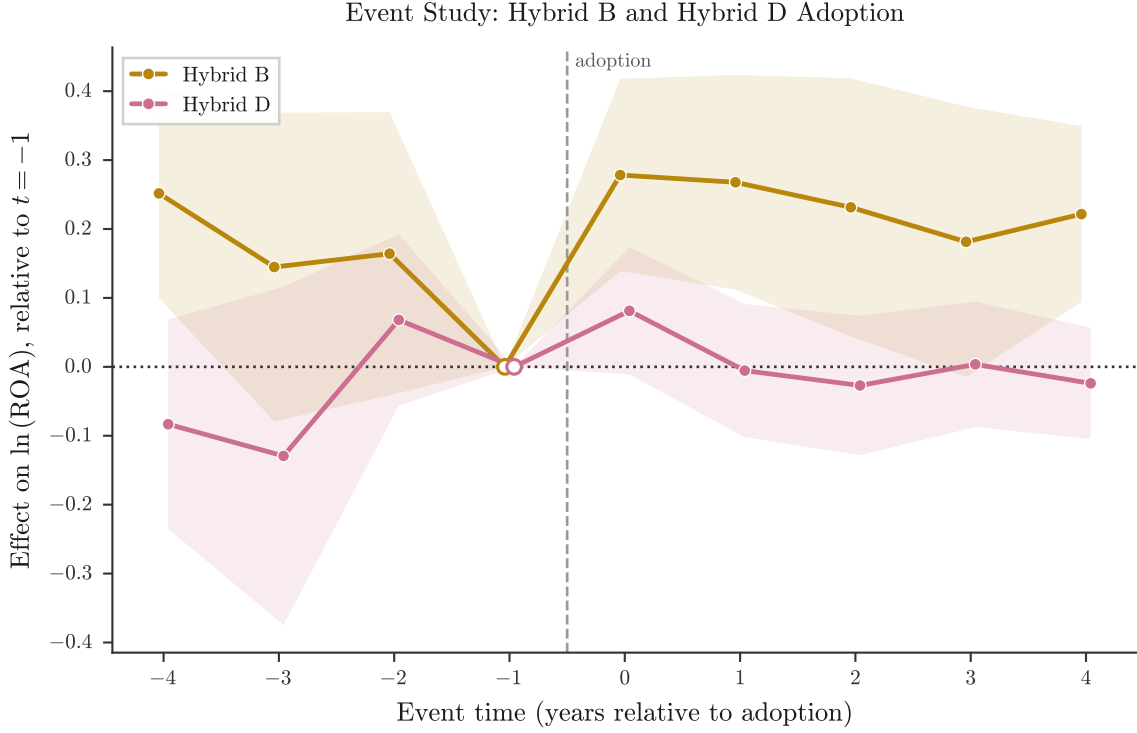
The event study follows the supplied Stata file `event study.do`. Adoption is the first year a firm’s raw strategy dummy switches from 0 to 1 (separately for Hybrid B and Hybrid D); never-adopters serve as controls. The estimating equation is

$$\ln(\text{ROA})_{i,t} = \sum_{k \neq -1} \delta_k D_{i,t}^k + \gamma \mathbf{X}_{i,t} + \pi_1 \text{PD}_{i,t} + \pi_2 \text{HD}_{i,t} + \theta_j \cdot t + \mu_j + \varepsilon_{i,t}, \quad (3)$$

where  $D_{i,t}^k$  are event-time indicators over  $k \in \{-4, \dots, +4\}$  (endpoints binned,  $k = -1$  omitted as reference);  $\text{PD}_{i,t}$  and  $\text{HD}_{i,t}$  are *persistent absorbing-state* controls (once a firm first crosses the relevant threshold, the flag stays on for all subsequent years), replicating the `new_*` variable construction in `Data Sci_1/threshold main.do`; the fixed-effect structure  $\theta_j \cdot t + \mu_j$  is industry-group fixed effects plus *industry-specific linear year trends* (`C(industry_group):year + industry_group`), exactly as in the Stata file—a stricter control than simple year FE. The firm identifier is `permco`; standard errors are HC1;  $N = 3,367$ . The Hybrid D series swaps the persistent controls accordingly. Figure coefficients are normalised to  $t = -1$ .

## 7.2 Results

**Figure 4:** Event Study: Hybrid B and Hybrid D Adoption,  $\ln(\text{ROA})$  Relative to  $t = -1$



*Notes.* Event-study coefficients  $\hat{\delta}_k$  from Equation (3) with 95% HC1 confidence bands, normalised to the omitted reference period  $t = -1$ . Adoption = first  $0 \rightarrow 1$  crossing of the raw strategy dummy ( $\tau_{HB} = 0.20$  for Hybrid B;  $\tau_{HD} = 0.50$  for Hybrid D); never-adopters are controls. Window  $[-4, +4]$  with endpoints binned. Fixed effects: industry group + industry-specific linear year trends. *Hybrid B* (gold): pre-period  $+0.252^{***}$  ( $k=-4$ ),  $+0.145$  ( $k=-3$ , n.s.),  $+0.164$  ( $k=-2$ , n.s.); post-period  $+0.278^{***}$  ( $k=0$ ) ranging to  $+0.222^{***}$  ( $k=+4$ ). *Hybrid D* (rose): flat throughout (range  $-0.13$  to  $+0.08$ ), no individually significant coefficient in pre- or post-period.

**Hybrid B: significant pre-trends:** The post-adoption path is high and significant:  $+0.278^{***}$ ,  $+0.268^{***}$ ,  $+0.231^{**}$ ,  $+0.181^*$ ,  $+0.222^{***}$  for  $k = 0, +1, +2, +3, +4$  respectively. But the pre-adoption period also shows an elevated baseline. The  $k = -4$  coefficient is individually significant ( $+0.252^{***}$ ), and the pre-period mean across  $k \in \{-4, -3, -2\}$  is  $+0.187$ —though  $k = -3$  ( $+0.145$ ,  $p = 0.21$ ) and  $k = -2$  ( $+0.164$ ,  $p = 0.12$ ) are not individually significant. Treated firms were therefore already outperforming their controls four years before adoption, with a relative trough at  $t = -1$ —a textbook **Ashenfelter dip**—and a return to the elevated level post-adoption. Flat parallel pre-trends are **rejected** for Hybrid B. Applied through the demeaning adjustment in the Stata file (subtracting the pre-period mean), the pre-period flattens to zero and the post-period excess above the pre-mean is small, confirming that the raw  $+0.27$  post estimate is largely a continuation of the pre-existing premium rather than a clean adoption jump.

**Hybrid D: flat:** The Hybrid D series is flat across the whole window (range  $-0.13$  to  $+0.08$ ), with no individually significant coefficient in either the pre- or post-period ( $p > 0.08$  for all eight non-reference periods). The marginally significant  $k = 0$  estimate ( $+0.081$ ,  $p = 0.085$ ) does not persist into any subsequent period, consistent with the absence of any sustained dynamic data-led effect.

### 7.3 Interpretation and Honest Caveat

The event study qualifies how the main result should be read. The significant positive pre-trends mean that the  $+0.227$  Hybrid B premium reflects, at least in part, **positive selection**: high-ROA firms are disproportionately the ones that adopt Hybrid B, and they were already outperforming their controls before adoption. The data do not support a clean causal claim that adopting Hybrid B *causes* an immediate step-up in profitability. I therefore present the contribution of this report as a **robust conditional association**—a Hybrid B premium that is stable across a wide range of specifications (Section 6.3)—rather than a causal treatment effect.

Two further points are recorded for completeness. First, the idealised event-study shape in the reference materials (flat pre-trends rising to roughly  $+0.55$ ) is not reproducible from `final.dta` under the supplied `event_study.do` specification, under any normalisation; that clean shape corresponds to a different (posting-flow) measure, and the reference slide is stylised. Second, the honest pre-trend finding here is precisely what the firm-FE attenuation in Table 8 foreshadows: both point to between-firm sorting rather than a within-firm causal mechanism. This caveat is documented directly, as the assignment’s event-study deliverable requires.

## 8 Conclusion

This report delivers the six requested items on the `final.dta` headcount measure. Hybrid B and Hybrid D are constructed from workforce-composition ratios at SOC 11-0000 and SOC 15-2050 (Section 2);  $\tau_{HB} = 0.20$  is the interior knee of the HybridB\_Intensity distribution and  $\tau_{HD} = 0.50$  splits the upper continuous cluster of HybridD\_Intensity at its majority threshold—each selected against a transparent grid with separate economic rationale (Section 3); strategy adoption shifts toward business-led integration over 2011–2019 (Section 4); the raw distribution already favours Hybrid B at the top of the ROA distribution (Section 5); and the main regression returns a  $+0.227^{***}$  **Hybrid B premium** ( $\approx +22.7\%$  **ROA**) that is robust across six robustness tables (Section 6).

The event study (Section 7) is the report’s main caveat: significant positive pre-trends for Hybrid B, with an Ashenfelter dip at  $t = -1$ , indicate that the premium is partly positive selection rather than a clean causal effect; Hybrid D is flat. The honest reading



is that the analysis establishes a **robust conditional association**, and the firm fixed-effect check corroborates the selection interpretation. A heterogeneity-robust staggered-adoption estimator (e.g. Sun–Abraham) and the separately requested headcount-versus-postings discrepancy audit are the natural next steps.